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EXP 10 – Create a data visualization (e.g., pie charts, bar graphs) for a Food Order Analysis using JavaScript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for analyzing food order data using JavaScript.

PROCEDURE:

Step 1: Set Up HTML File

Create an HTML file to include canvas elements for charts and link Chart.js library.

Step 2: Create JavaScript File

Create a JavaScript file to define food order data and generate charts.

Step 3: Define Data

Add labels such as Pizza, Burger, Pasta, Drinks, and Desserts with corresponding order values.

Step 4: Create Charts

Use Chart.js to create:

- Bar chart for comparison
- Pie chart for distribution

Step 5: Run the Program

Open the HTML file in a browser to view the charts.

PROGRAM:

HTML CODE:

index.html × script.js

index.html > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Food Order Analysis</title>

<style>
body {
font-family: Arial, sans-serif;
text-align: center;
background: #f7f9fc;
margin: 40px;
}

h1 {
color: #2c3e50;
margin-bottom: 30px;
}

.chart-box {
background: white;
padding: 20px;
margin: 25px auto;
border-radius: 15px;
box-shadow: 0 4px 12px rgba(0,0,0,0.1);
width: 420px;
}

canvas {
width: 100% !important;
}
</style>

</head>

<body>
```

```
margin: 25px auto;
border-radius: 15px;
box-shadow: 0 4px 12px rgba(0,0,0,0.1);
width: 420px;
}

canvas {
width: 100% !important;
}
</style>

</head>

<body>

<h1>Food Order Analysis Dashboard</h1>

<div class="chart-box">
<canvas id="barChart"></canvas>
</div>

<div class="chart-box">
<canvas id="pieChart"></canvas>
</div>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>

</body>
</html>
```

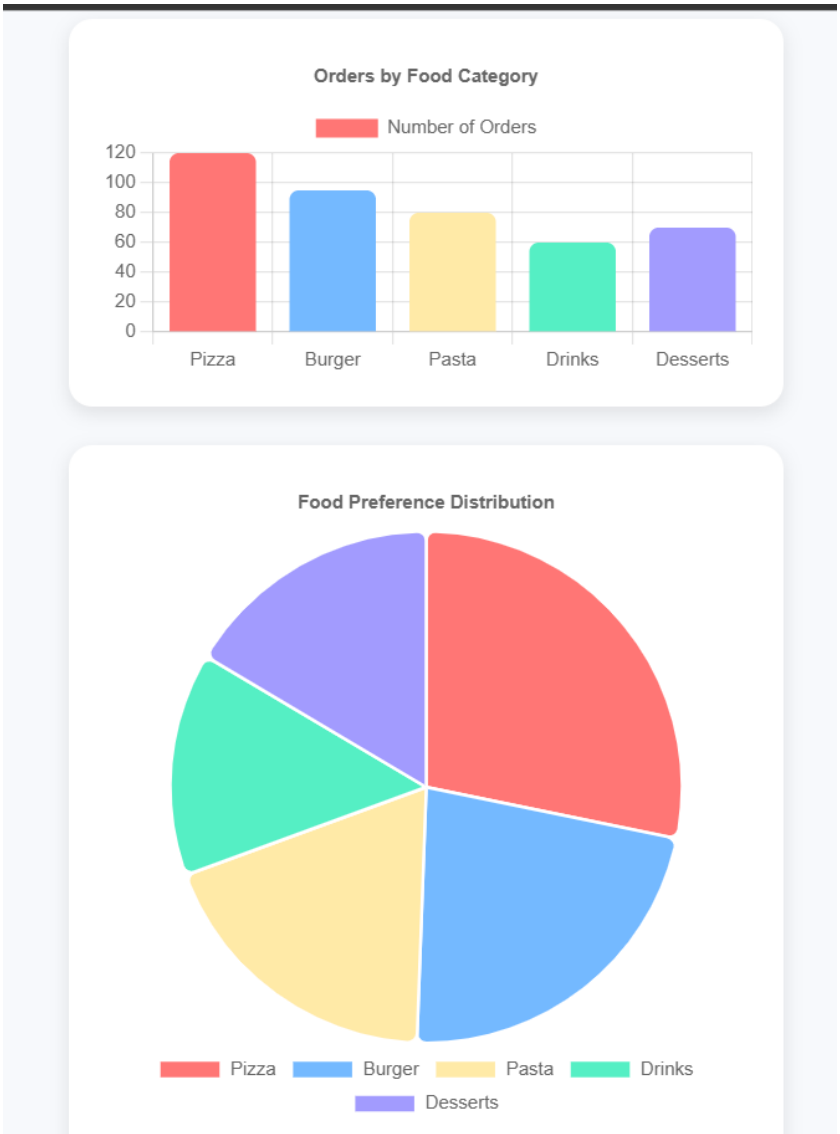
JAVASCRIPT CODE:

```js
// Food Order Data
const foodData = {
labels: ['Pizza', 'Burger', 'Pasta', 'Drinks', 'Desserts'],
datasets: [{
label: 'Number of Orders',
data: [120, 95, 80, 60, 70],
backgroundColor: [
'#ff7675',
'#74b9ff',
'#ffeaa7',
'#55efc4',
'#a29bfe'
],
borderRadius: 6
}]
};

// BAR CHART
new Chart(document.getElementById('barChart'), {
type: 'bar',
data: foodData,
options: {
responsive: true,
plugins: {
title: {
display: true,
text: 'Orders by Food Category'
}
},
scales: {
y: {
beginAtZero: true
}
}
}
});
```

```
// PIE CHART
new Chart(document.getElementById('pieChart'), {
type: 'pie',
data: foodData,
options: {
responsive: true,
plugins: {
title: {
display: true,
text: 'Food Preference Distribution'
},
legend: {
position: 'bottom'
}
}
}
});
```

OUTPUT:



RESULT:

Thus, data visualization for food order analysis using pie and bar charts was successfully implemented using JavaScript.